

LOW-COST STOPWATCH

■ PRADEEP G.

Here is a simple circuit of a stopwatch that can count up to 99 seconds. The circuit uses CD4060, CD4013, CD4033, LTS543 and some discrete components. A crystal-controlled oscillator generates 1Hz pulses. A 32.768kHz miniature crystal is used for the purpose.

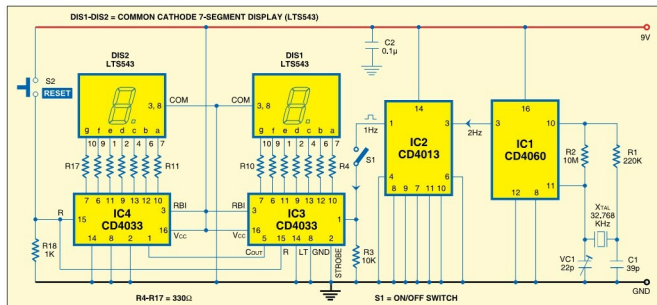
IC CD4060 (IC1) contains a 14-bit divider stage and oscillator section. The 2Hz output of IC1 at its pin 3 is fed to flip-flop IC CD4013 (IC2). IC2 converts 2 Hz into 1 Hz. The output of

IC2 at pin 1 is fed to IC CD4033 (IC3) through switch S1. Carry-out pin 5 of IC3 is connected to pin 1 of another CD4033 (IC4) for further counting. IC3 and IC4 drive common-cathode displays DIS1 and DIS2 to show the time (maximum 99 seconds). Switch S1 is used to start and stop the watch. You can reset the counter to zero by pressing reset switch S2.

For proper operation, first connect 9V power supply to the circuit, then reset the display by momentarily pressing switch S2. The display shows '00.' Now to start counting, flip switch S1 to

'on' position. At the time of completion of event, flip switch S1 to 'off' position. Time shown on the display is the time for the event.

Assemble the circuit on a general-purpose PCB and enclose in a suitable case. Mount 7-segment, common-cathode displays DIS1 and DIS2, and switches S1 and S2, on the front panel of the cabinet. As the circuit employs CMOS ICs, supply voltage may vary from 5V to 15V. 9V supply is recommended. ●



REMOTE-OPERATED MASTER SWITCH

■ D. MOHAN KUMAR

Generally, a bedside master switch is used to switch on lamps both indoors and outdoors when there is a threat of intruder. This circuit can be used to activate the master switch from the bed without searching for the switch in darkness. It can be activated by the TV remote handset. The security lamps

glow for three minutes and then turn off. The circuit is sensitive and can be activated from a distance of up to 25 metres.

IR receiver module TSOP 1738 (IRX1) is used to sense the pulsed 38kHz IR rays from the TV remote handset. The IR receiver module has a PIN photodiode and a preamplifier enclosed in an IR filter epoxy case. Its open-collector output is 5 volts at 5mA

current in the standby mode.

In the standby mode, no IR rays from the remote handset fall on the IR receiver, so its output pin 3 remains high and LED1 doesn't glow. Through